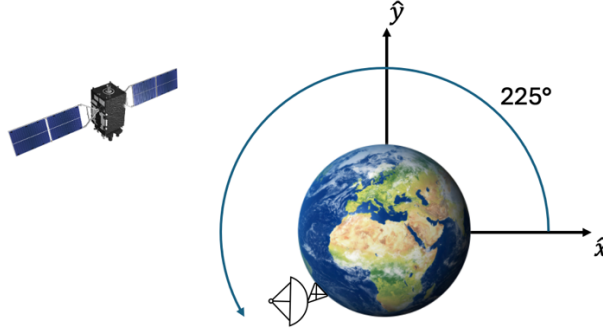


Space Mission and System 2025/2026 – Challenge

During the commissioning phase, an Earth orbiter is tracked by a ground station providing range and range-rate observables with an expected accuracy of 1 cm and 1 mm/s.



In addition to Earth gravity (consider only the monopole term), the spacecraft is subject also to the drag acceleration from the atmosphere, described by the usual formula:

$$\bar{a}_{drag} = -\frac{1}{2}\rho(z)C_D\frac{A}{m}V^2\hat{V}$$

Where ρ is the atmospheric density (in kg/m^3), z is the spacecraft altitude, $C_D = 2$ the drag coefficient, $A = 25 \text{ m}^2$ the spacecraft's cross-sectional area, $m = 1000 \text{ kg}$ the spacecraft's mass, V and $\hat{V}$ the velocity module and unit vector. Both the Earth and Earth atmosphere are assumed to be non-rotating. The atmosphere density law can be described as follow:

$$\rho(z) = \rho(z_0)e^{-\frac{z-z_0}{H}}$$

where $z_0 = 300 \text{ km}$ and $H = 14.7 \text{ km}$. The “a priori” values and uncertainties of the state vector components at the reference time $t_0 = 0$ are:

State vector components	Value	Uncertainty
x	$+4.7230591 \times 10^3 \text{ km}$	10 m
y	$+4.7210591 \times 10^3 \text{ km}$	10 m
V_x	-6.2288709 km/s	1 m/s
V_y	$+6.2283709 \text{ km/s}$	1 m/s
$\rho(z_0)$	$+1.8080000 \times 10^{-2} \text{ kg/m}^3$	0.1 kg/km^3

To recover Homework 1, answer the following questions:

- (1) Using both range and range-rate observables provided by the ground station (found in the file “observable.txt”), obtain an estimate of the five components of the state vector and the associated uncertainties at the reference epoch.
- (2) Which of the two types of observables is most valuable for the determination of the probe state? Justify your answer.
- (3) Considering a new estimation procedure, **including also the drag coefficient C_D in the estimated vector with an “a priori” value of 2.014 and an “a priori” accuracy of 0.01**, generate a new orbit determination solution, commenting the final results.

After commissioning, the spacecraft operations enter the nominal phase. Now the spacecraft navigates autonomously using an Extended Kalman filter (EKF) for orbit determination. In this scenario, the ground station tracking ends, and the navigation filter relies on the pseudo-range and pseudo-range rate observables from terrestrial GNSS satellites. The spacecraft collects pseudo-range and pseudo-range rate measurements from the E11 satellite of the Galileo constellation with a putative accuracy of 10 cm and 1 cm/s, whose orbit can be described in the X-Y plane using the following Keplerian orbit parameters:

Keplerian Parameters	Value
Semi-major axis: a	29599.8 km
Right Ascension: Ω	77.632 deg
Argument of the pericentre: ω	0.0 deg
Eccentricity: e	0
True Anomaly at the epoch: ν_0	15.153 deg

The observable functions for the pseudo-range and pseudo range-rate observables can be described as follow:

$$\rho_{PR} = \sqrt{(X_{sc} - X_{GNSS})^2 + (Y_{sc} - Y_{GNSS})^2} + c\delta t$$

$$\dot{\rho}_{PR} = \frac{(X_{sc} - X_{GNSS})(V_{x,sc} - V_{x,GNSS}) + (Y_{sc} - Y_{GNSS})(V_{y,sc} - V_{y,GNSS})}{\sqrt{(X_{sc} - X_{GNSS})^2 + (Y_{sc} - Y_{GNSS})^2}} + c\dot{\delta t}$$

Where δt and $\dot{\delta t}$ are the unknown bias and drift rate of the spacecraft's clock and c is the speed of light. The two quantities are related in the following way:

$$\delta t = \delta t_0(t = t_{0,GNSS}) + \dot{\delta t} \cdot t$$

Without changing the dynamical model, solve the following points using an EKF:

- (C1) Augmenting the original estimation vector (without C_D) by adding the clock bias δt and the clock drift $\dot{\delta t}$, generate a new estimate using the EKF. Plot the estimated state vector components as a function of time. In addition, plot also the residual vs. time for both observables. The pseudo-range and pseudo-range-rate observables are found in "observables_gnss.txt". **IMPORTANT: Use as "a priori" information the results of the previous point and consider for the new parameter the following values: $\delta t_0 = 2 \times 10^{-6} s$ and $\dot{\delta t} = 2.1 \times 10^{-8} s/s$, with "a priori" accuracy of 1 s and 0.001 s/s respectively.**
- (C2) Once the full timeline of the clock bias is obtained, find the new value at t_0 , hence the value of δt_0 , by means of a linear regression.
- (C3) Plot the trace of the covariance vs. time.

NOTE: For the estimation procedure of point (C1) of the challenge use the full covariance matrix of the last iteration of point (1).

Read carefully the following information:

- Earth gravitational parameter (GM): $3.986004418 \times 10^5 \text{ km}^3/\text{s}^2$
- Earth radius: 6378 km
- Speed of light (c): 299792.458 km/s
- $t_{0,GNSS} = 16100 \text{ s}$

Upload on Google Classroom a **zip folder** containing a working computer code and instructions for its use. The code must run with a one-line command. It must print tables and produce the relevant plots (MATLAB is recommended). You must provide a **comprehensive and concise note in English** (a pdf file **not included in the zip folder**) containing **all** mathematical procedures, results, comments, tables and figures. **Deadline: Sunday 26 April 23:59:59. No exceptions.**

Only questions pertaining to the text will be answered. Please ask questions publicly on Classroom (in English).

Tips on MATLAB:

- Rather than ode45, use ode113, which excels in orbital dynamics problems (just replace ode45 with ode113 in your setup);
- For integration purposes, use at least RelTol=1.0E-13, AbsTol=1.0E-13;
- ALWAYS put labels and units on the plot axes!